

Assignment-3

Day 5: Case Study Practical Work on a scenario involving multiple hardware issues.

Task-1: A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step 1: Check the wall outlet

- Plug another working device into the same outlet.
- **Checking for:** Whether the power outlet is actually supplying electricity.

Step 2: Inspect the charger

- Check the adapter, cable, and connector for cuts, burns, bent pins, or loose connections.
- **Checking for:** Physical damage to the charger.

Step 3: Check the charging indicator

- Connect the charger and look for the laptop's charging/power LED.
- **Checking for:** Whether the laptop is receiving power from the adapter.

Step 4: Perform a basic power reset

- Disconnect the charger.
- If the battery is removable, remove it.
- Hold the power button for about **15–20 seconds**.
- Reconnect the charger and try powering on.
- **Checking for:** A temporary power-management or residual-charge problem.

Step 5: Test the charger/battery

- If available, test with a **known-compatible charger** with the correct voltage and power rating.
- If the laptop works with the replacement charger, the original charger may be faulty.
- If there is still no response, the problem could involve the **battery, charging port, motherboard, or power circuitry**.

Task-2: You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Troubleshooting Flow

1. Check the monitor power

- Confirm that the monitor is switched on.
- Check its power cable and power indicator.

2. Check the video cable

- Make sure the HDMI/DisplayPort cable is securely connected at both ends.
- Check for damaged cables or connectors.

3. Check the correct input source

- Use the monitor's buttons to select the correct HDMI/DisplayPort input.

4. Check where the cable is connected

- If the PC has a dedicated GPU, connect the monitor cable to the **GPU's video output**, not the motherboard output.

5. Test another cable or monitor

- Use a known-working cable.
- If possible, connect another monitor to the PC.

6. Check the GPU

- Shut down the PC and disconnect power.
- Make sure the GPU is properly seated and its power connectors are attached.
- Check that its fans and other indicators behave normally.

7. Check RAM/POST

- If there is still no display, check for motherboard diagnostic LEDs or POST beep codes.
- Reseat the RAM and test one module at a time if necessary.

8. Confirm the solution

- Start the PC and verify that the BIOS/UEFI screen and operating system display correctly.

Task-3: A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

I would isolate the problem by testing the **mouse** → **USB port/receiver** → **system** in that order.

Step 1: Check the mouse

- Check whether the mouse is switched on.
- Replace or recharge the batteries.
- Check whether the optical sensor is clean.
- Test the mouse on another computer.

Result: If it doesn't work on another computer either, the mouse is probably faulty.

Step 2: Check the USB receiver/port

- Remove the wireless receiver and reconnect it.
- Try a **different USB port**.
- Check whether other USB devices work in the original port.

Result: If the mouse works in another USB port, the original USB port may have a problem.

Step 3: Check system settings

- Check whether Bluetooth/wireless connectivity is enabled if the mouse uses Bluetooth.
- Check **Device Manager** for mouse/HID or USB errors.
- Restart the computer and test again.

Result: If the mouse works after changing settings or reconnecting the receiver, the issue was likely related to the system configuration.

Task-4: Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Possible Cause 1: CPU/GPU overheating

- **Check:** Open the case and inspect the CPU and GPU temperatures using the BIOS hardware monitor or appropriate diagnostic tools.
- **Look for:** Very high temperatures, fans running at maximum speed, or thermal shutdown during gaming.
- **Action:** Clean dust from the heatsinks and fans and make sure the cooling fans are working correctly.

Possible Cause 2: Dust or blocked airflow

- **Check:** Inspect the CPU heatsink, GPU heatsink, case fans, and air vents.
- **Look for:** Heavy dust buildup blocking airflow.
- **Action:** Power off and unplug the PC, then clean the dust using appropriate compressed air and ensure the case has adequate airflow.

Possible Cause 3: Faulty or poorly functioning cooling fan

- **Check:** Turn on the PC and observe whether the CPU, GPU, and case fans spin normally.
- **Look for:** A fan that does not spin, makes unusual noises, or stops intermittently.
- **Action:** Check the fan connection and replace a faulty fan if necessary.